

Multiple Choice (10 marks)

1. In Python 3, what is the following function returns item from the list with max value?
 - (a) `cmp(list)`
 - (b) `len(list)`
 - (c) `max(list)`
 - (d) `min(list)`
2. Which of the following has the greatest potential for compromising a user's personal privacy?
 - (a) A group of cookies stored by the user's Web browser
 - (b) The Internet Protocol (IP) address of the user's computer
 - (c) The user's e-mail address
 - (d) The user's public key used for encryption
3. Which pillar of cybersecurity is compromised when someone logs into a system using a stolen login and password?
 - (a) Authentication
 - (b) Confidentiality
 - (c) Integrity
 - (d) Nonrepudiation
4. Let `l = [1,2,3,4]`. What is `min(l)` in Python 3?
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
5. What is the value of this python expression: `1 + 3 % 3`?
 - (a) 0
 - (b) 1
 - (c) 3
 - (d) 4

True / False (10 marks)

Answer True or False

6. True or False: $O(N^{1/4})$ grows slower than $O(N^{1/N})$ for large values of N .
7. True or False: $O(1)$ is the smallest asymptotic notation, indicating constant time complexity regardless of input size.
8. True or False: A large Java program that has been tested may still contain hidden bugs despite no errors being found during testing.
9. True or False: A denial-of-service (DoS) attack always targets just one computer at a time, unlike a DDoS attack.
10. True or False: The output of the statement `'a' + 'ab'` in Python 3 is `'aab'`.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Explain how you would implement the `nextIntInRange` method to ensure it returns a random integer between two specified values, low and high, inclusive. Discuss your reasoning.
12. Explain the process that occurs when a user requests a file from a web server, focusing on how the file is transmitted and the importance of packet reassembly upon receipt.

End of Paper